# Hyunho Lee

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## Research Interests

Remote sensing images based deep learning model for the water resources management

Spatiotemporal data mining for the water resources management

Spatial data science and geographic information science

## Education

**Ph.D. Geographic Information Science** — **Aug. 2022 – Present**
*Arizona State University* — *Tempe, United States*
Advisor: Prof. Wenwen Li

**M.S. Computer Science** — **Mar. 2008 – Feb. 2010**
*KAIST (Korea Advanced Institute of Science and Technology)* — *Daejeon, Republic of Korea*
Thesis: The layer-based vector texture for 3D rendering
Advisor: Prof. Kwangyun Wohn

**B.E. Information and Computer Engineering** — **Mar. 2001 – Aug. 2007**
*Ajou University* — *Suwon, Republic of Korea*

## Publications

[1] J Park, **H Lee** (2020). Prediction of high turbidity in rivers using LSTM algorithm. *Journal of Korean Society of Water and Wastewater*, 34(1), 35-43. https://doi.org/10.11001/jksww.2020.34.1.035

[2] J Park, **H Lee**, CY Park, S Hasan, TY Heo, WH Lee (2019). Algal morphological identification in watersheds for drinking water supply using neural architecture search for convolutional neural network. *Water*, 11(7), 1338. https://doi.org/10.3390/w11071338

[3] M Park, Y Yoon, **H Lee**, J Kim (2018). Application of recurrent neural network for inflow prediction into multi-purpose dam basin. *Journal of Korea Water Resources Association*, 51(12), 1217-1227. https://doi.org/10.3741/JKWRA.2018.51.12.1217

## Work Experience

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|---|---|
| **Senior Manager**, Digital Water Platform Dept., *K-water* | Jan. 2021 – Jun. 2022 |
| **Manager**, Digital Innovation Dept., *K-water* | Jan. 2020 – Dec. 2020 |
| **Manager**, Data Center Dept., *K-water* | Jan. 2019 – Dec. 2019 |
| **Manager**, Water Data Collection and Analysis Dept., *K-water* | Jan. 2018 – Dec. 2018 |
| **Manager**, Human Resources Management Dept., *K-water* | Jan. 2013 – Dec. 2017 |
| **Staff**, Information System Management Dept., *K-water* | Jul. 2010 – Dec. 2012 |

## Teaching Experience

**In-house Instructor**, Data Analysis with R programming, *K-water* Apr.2018 – May.2021

**Teaching Assistant**, Introduction to Programming (CS101), *KAIST* Mar. 2009 – Jul.2009

## Honors and Awards

**1st Place Prize**, 5th Bigdata analysis competition, *K-water* Oct. 2021

**Academic Conference Paper Award**, Korean Society of Environmental Engineering Annual Conference Nov. 2020

**12th place**, ACM-ICPC (International Collegiate Programming Contest) Asia-Seoul Regional Nov. 2003

## Certification

**Advanced Data Analytics Professional**, *certificated by K-Data, Republic of Korea* Apr. 2019
(Pass rate: 2.76%)

## MOOC's – Online Certification

**Machine Learning**, CSMM.102x, *ColumbiaX (EDX)* Jan. 2017 – Apr. 2017

**Mathematical Fundamentals for Data Science**, *Korea University (K-MOOC)* Oct. 2017 – Jan. 2018

**Artificial Intelligence and Machine Learning**, *KAIST (K-MOOC)* Jan. 2018 – Feb. 2018

## Soft Skills

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| **Programming Languages** | Python, R, C/C++, JAVA, ABAP (SAP) |
| **Data Science and Machine Learning** | PyTorch, Keras, Tensorflow, Sci-kit Learn |
| **Visualization** | Matplotlib, Plotly, Leaflet, QGIS, OpenGL |